MOEMS®
Mathematical Olympiads for Elementary and Middle Schools

PROBLEM OF THE WEEK

2025-2026

SOLUTIONS

We do not “Share” this file.
To make a copy go to: File > Make a Copy

MOEMS®

Mathematical Olympiads for Elementary and Middle Schools

6/29/26

Strategy: Convert all the times to London time.
From the first sentence, London time (LT) is 5 hours ahead of New York time (NYT). The first plane departs at 12 noon LT and arrives at 11 AM NYT, which is 4 PM LT, so the travel time is 4 hours. The second plane leaves New York at 12 noon NYT, which is 5 PM LT. The travel time is 4 hours,

so it is 9 pm in London when the second plane arrives.

MOEMS®
Mathematical Olympiads for Elementary and Middle Schools

6/22/26

The largest amount that can be made is 49¢. Using the given set of coins, any amount from 1¢ to 49¢ can be made.

Therefore, there are 49 different amounts that can be made.

MOEMS®
Mathematical Olympiads for Elementary and Middle Schools

6/15/26

The answer to this problem is the number of different pairs that can be formed with the four students.Denote the different students by A, B, C, and D. Then the different pairs are: AB, AC, AD, BC, BD, and CD.

Since there are 6 different pairs, then there are 6 days when it can be arranged that the same pair of students do not work together. (Note that AB and BA stand for the same pair; AB and BC are different pairs.)

MOEMS®

Mathematical Olympiads for Elementary and Middle Schools

6/8/26

dimension 1:	1	2	3	4	5	6
dimension 2:	11	10	9	8	7	6
area:	11	20	27	32	35	36

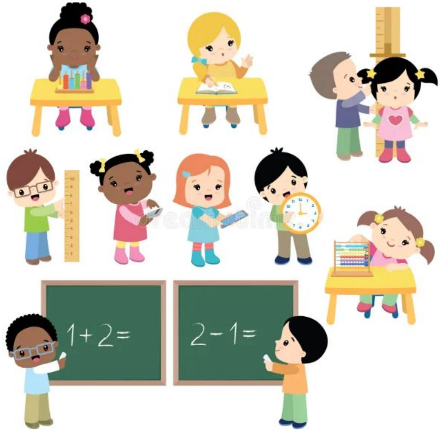
The semiperimeter of the rectangle is 12 meters. Use a table to show some possible dimensions in meters and the corresponding areas in square meters.

According to the dimensions shown, the largest area is 36.

MOEMS®
Mathematical Olympiads for
Elementary and Middle Schools

6/1/26

number of week:	1	2	3	4	5
number of new members:	1	2	4	8	16



Total of new members: 1+2+4+8+16=31.

Volume 1, Olympiad 35, #4, 5 Minutes

MOEMS®
Mathematical Olympiads for Elementary and Middle Schools

5/18/26

In 10 minutes, A travels 7,000 yards, B travels 8,000 yards, and C travels 9,000 yards.

Therefore, at 10 minutes, each is at the starting point and together again for the first time since they started the race.

MOEMS®

Mathematical Olympiads for Elementary and Middle Schools

5/18/26

1		7	12
	4	9	
	5	16	
8	11		13

The sum of the 4 entries on the major diagonal is 34. Then the missing entry at the bottom of the thirdcolumn should be 2. Calculating B produces B = 13. Similarly, A + 4 + 16 + 13 = 34 or A + 33 = 34.

Therefore A = 1 and B = 13.

Volume 1, Olympiad 40, #2. 5 Minutes

MOEMS®
Mathematical Olympiads for Elementary and Middle Schools

5/11/26

Strategy: Count paths to each letter separately.

1. From O, there are 2 possible paths to L.
2. From each L, there are 2 possible paths that lead to a Y (for example, L_1Y_1 and L_1Y_2), so the total number of paths from O to Y is: $2 \times 2 = 4$
3. From each Y, there is only 1 path leading to M, so the total number of paths from O to M remains: $4 \times 1 = 4$
4. Next, from each M, there are 2 possible paths leading to a P, giving a total of: $4 \times 2 = 8$ paths from O to P.
5. From each P, there are 3 possible paths leading to an I, so the total number of paths from O to I is: $8 \times 3 = 24$
6. Then, from each I, there are 2 possible paths leading to an A, resulting in: $24 \times 2 = 48$ paths from O to A.
7. Finally, from each A, there is only 1 path leading to D.

Therefore, the total number of paths from O to D is: $48 \times 1 = 48$

MOEMS®
Mathematical Olympiads for
Elementary and Middle Schools

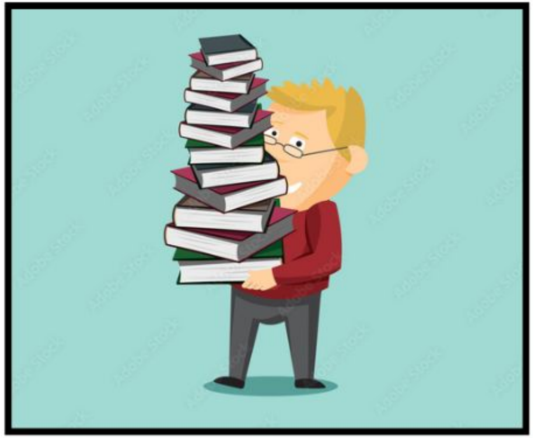
5/4/26

Strategy: Consider how frequently both lights flash together.
From 1:00 to 3:00 is 120 minutes. We need the least multiple of 14 that is greater than 120. 120 divided by 14 is between 8 and 9. Then $9 \times 14 = 126$ is the least such multiple of 14. 126 minutes after 1:00 PM is 3:06 PM.

The first time they flash together after 3 PM is 3:06 PM

MOEMS®
Mathematical Olympiads for Elementary and Middle Schools

4/27/26



Strategy: Combine their work into a 2-day total, then scale to reach 500.
The man binds 100 books and his helper binds 25 books for a total of 125 books every 2 days. Thus, binding 500 books requires 4 sets of 2 days each.

It would take 8 days to bind 500 books.

Volume 2, Set 5, Olympiad 5, 5B. 4 Minutes

MOEMS®
Mathematical Olympiads for Elementary and Middle Schools

4/20/26

Strategy: A group of 3 combined with a group of 2 makes a group of 5. Think of the class as consisting of 5 groups of equal size: 3 groups vote for Kim and 2 for Amy. After 8 people switch, think of the class as now consisting of 3 groups of equal size (each different in size from the original grouping): 2 groups vote for Amy and the remaining group votes for Kim. To be divided into either 3 equal groups or 5 equal groups, the class must contain 15, 30, 45, or any multiple of 15 people. Suppose 15 people are in the class. Then Kim would win, 9–6. Switching 8 votes gives Amy a 14–1 win. Reject — the ratio is not 2:1.
Suppose 30 people are in the class. Then Kim would win, 18–12.
Switching 8 votes gives Amy a 20–10 win. Accept — the ratio is 2:1.

Thus a total of 30 people voted.

MOEMS®
Mathematical Olympiads for Elementary and Middle Schools

4/13/26

Strategy: Use the meaning of percent.
The fraction of all adults starting a new business last year was 1/N and this year it was 1/55. An increase of 20% means that the new rate is 120% of the old rate, or 1.2 times the old rate.

The old rate multiplied by 1.2 is the new rate: $(1.2)\frac{1}{N} = \frac{1}{55}$

Multiply 1/N by 1.2: $\frac{1.2}{N} = \frac{1}{55}$

Cross-Multiply: $(N)(1) = (1.2)(55)$

Simplify: $N = 66$

N=66

NOTE: The equation can be solved in several other ways but the result will always be N = 66

MOEMS®

Mathematical Olympiads for Elementary and Middle Schools

4/6/26

Strategy: Use a frequency definition of probability.
Use a frequency definition of probability. Consider a large and convenient number of days, say 100. Rain is expected for 40 days and fair weather for 60 days. Jess would expect to earn a total of (40 × \$1500) + (60 × \$400) = \$84,000. During the 100 day period,

Jess expects to earn \$84,000, which is an average of \$840 daily.

MOEMS®
Mathematical Olympiads for Elementary and Middle Schools

3/30/26

Strategy: Start with a different number of people in each car.
If the number of people in every car is different and no car is empty, the minimum number of
people would be 1 + 2 + 3 + 4 + 5 = 15. But there are only 12 people, and no car has 5 people. Take 3 of the 5 people out of that last car, leaving 1 + 2 + 3 + 4 + 2 = 12 people.

Only the two red cars have the same number, so each red car has 2 people.

MOEMS®
Mathematical Olympiads for Elementary and Middle Schools

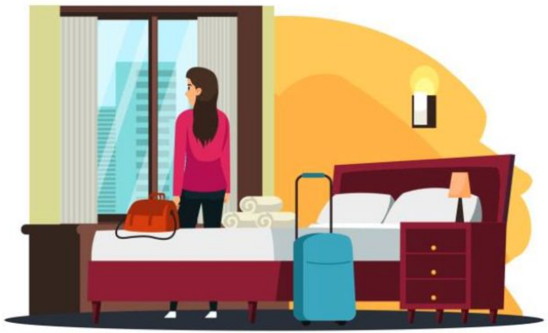
3/23/26

Strategy: Use the algorithms for addition and multiplication.
Since zero is never a leading digit, the letter O does not represent zero.
In the addition, O + O does not produce a "carry", so O = 1,2, 3, or 4.
In the multiplication, 4 × O + a possible "carry" of 0, 1, 2, or 3 gives the two-digit number ending in FO.
Suppose O= 1. Then 4 × 1 + the carry = F1. No carry works.
Suppose O = 2. Then 4 × 2 + the carry = F2. No carry works.
Suppose O= 3. Then 4 x 3 + the carry = F3. This works if the carry is 1.
Suppose O = 4. Then 4 × 4 + the carry = F4. No carry works.

The letter O represents the digit 3.

MOEMS®
Mathematical Olympiads for Elementary and Middle Schools

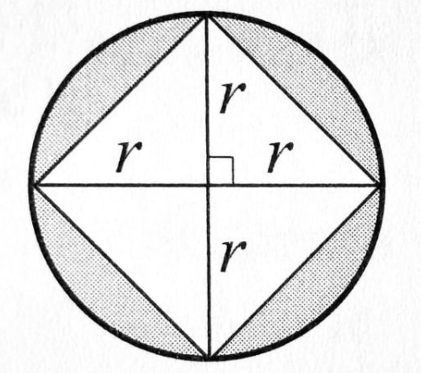
3/16/26



Strategy: Start with 1 person in each room.
After placing 26 people, 1 per room, there are 14 people left over.
Place these 14 people, 1 per room, so that there are 14 rooms with 2 people in each.

That leaves 12 rooms occupied by exactly one person.

3/9/26



Strategy: Draw some useful line segments.
The area of the shaded region is the difference between the areas of the circle and the square. Draw the diagonals of the square. The diagonals split the square into four congruent isosceles right triangles. The area of each triangle is $18 \div 4 = 4.5$.
The diagonals are also diameters of the circle, so the base and height of each of the small right triangles is a radius of the circle.
The area of each triangle, $\frac{1}{2} \times r \times r$ is 4.5, so $r = 3$ cm. The area of the circle is approximately $3.14 \times 3^2 = 28.26$.
The difference between the areas of the circle and square is $28.26 - 18 = 10.26$.

Rounding off, the area of the shaded region is 10 sq cm.

MOEMS®
Mathematical Olympiads for Elementary and Middle Schools

3/2/26

Strategy: Determine the number of letters in the repeating part. In the sequence ABBCCDABBCCD..., there are 6 letters before the pattern repeats itself. The first 78 terms of the sequence contain 13 complete repetitions of the pattern and no additional letters.

The 78th letter is the same as the 6th letter, which is D.

Volume 3, Set 7, Olympiad 2, 2B. 4 Minutes

MOEMS®

Mathematical Olympiads for Elementary and Middle Schools

2/26/26

985

Strategy: Determine which digits satisfy each condition.
The single digit prime numbers are 2, 3, 5, and 7. The single digit perfect squares are 0, 1, 4, and 9, leaving 6 and 8 as neither prime nor perfect square. Choose the largest digit in each group and arrange from greatest to least to get the largest possible number, 987. But 987 is divisible by 3. To get the next greatest number, choose the next greatest digit from the group containing 7.

The greatest possible value of Janine's number is 985.

MOEMS®
Mathematical Olympiads for Elementary and Middle Schools

2/16/26

NAME	ACTION	MUST HAVE STARTED WITH
Selena	Took 5, left 4	9 coins
Joe	Took 2, left 9	11
	Took half, left 11	22
Nick	Took 2, left 22	24
Sara	Took 4, left 24	28
	Took half, left 28	56

Strategy: Work backwards.
The table below shows actions in reverse. The first column names each person, last to first. The second column shows the actions each person took. The third column states the number of coins on the table as each person approached.

56 coins were on the table to start with.

MOEMS®
Mathematical Olympiads for
Elementary and Middle Schools

2/9/26

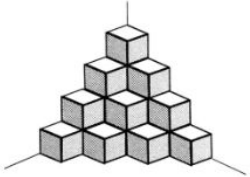
Strategy: Determine how many times Keri came in first.
Keri could not have finished first 3 or more times because that would give her more than 20 points. Suppose she finished first twice, a total of 16 points. Any remaining points must have come from finishing second, scoring 3 points each time. She can't have scored 4 points. The same reasoning shows she can't have finished first 0 times since 20 is not a multiple of 3. So Keri finished first once, for 8 points. The remaining 12 points came from 4 second place finishes.

The remaining 12 points came from 4 second place finishes.

Volume 3, Set 7, Olympiad 1, 1C. 5 Minutes

MOEMS®
Mathematical Olympiads for Elementary and Middle Schools

2/2/26



Layer	Cubes in layer (visible + hidden)
1 (top)	1 = 1
2	2 + 1 = 3
3	3 + 3 = 6
4 (bottom)	4 + 6 = 10

Strategy: Count horizontally, from the top layer down.
Make a table that counts cubes separately for each layer. In each case, add the number of hidden cubes to the number of visible cubes.

1 + 3 + 6 + 10 = 20 cubes are used to form the tower.

MOEMS®
Mathematical Olympiads for Elementary and Middle Schools

1/26/26

Strategy: Combine the two cases.
Suppose 2 students are absent and Mr. Alvarez still gives each student 4 sheets. He will have the original 16 sheets left over and in addition the 4 sheets that he would have given to each of the absentees. This total of 24 sheets is enough to give each of the students who are present 1 additional sheet, with 3 left over. Then there are 24 – 3 = 21 students present.

Mr. Alvarez has 5 × 21 + 3 = 108 sheets of paper.

MOEMS®
Mathematical Olympiads for Elementary and Middle Schools

1/19/26

Day Number	1	2	3	4	5	6	…	30
Quantity eaten this day	1	3	5	7	9	11	…	
Total eaten to date	1	4	9	16	25	36	…	900

Strategy: Start with the first few days; look for a pattern.
The total Lou eats for the first N days is N2. September has 30 days,

so Lou eats 900 jelly beans in September.

MOEMS®
Mathematical Olympiads for Elementary and Middle Schools

1/12/26

Strategy: Work backwards.
Ben ended with twice as many of the 36 CDs as Ali, so Ben ended with 24 and Ali with 12.
Ali had given Ben 40%, or 2/5 of her original number, so the 12 she ended with was 3/5 of the number she started with.

Then 1/5 of her original number was 4, and Ali had 20 CDs originally.

MOEMS®
Mathematical Olympiads for Elementary and Middle Schools

1/5/26

Strategy: Use counting principles.
Emma buys a German novel and a Spanish novel, or a German novel and a French novel, or a Spanish novel and a French novel. She has 4 choices for the German novel and 5 for the Spanish novel, so she can buy novels in those two languages in 4 x 5 = 20 ways. Likewise, she can buy a German novel and a French novel in 4 x 6 = 24 ways, and she can buy a Spanish novel and a French novel in 5 x 6 = 30 ways.

In all Emma can purchase two novels in two languages in 20 + 24 + 30 = 74 ways.

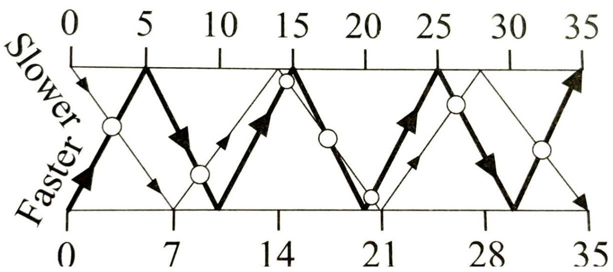
12/22/25

Strategy: Minimize the number of ushers.
Group each 30 fans with 1 usher to form groups of 31. Then the 20,000 people are divided into 645 groups of 31 each, with 5 people left over. Those 5 people must contain at least 1 usher and at most 4 fans. There must be at least 645 + 1 = 646 ushers.

There are at most 20,000 – 646 = 19,354 fans that can be in attendance.

MOEMS®
Mathematical Olympiads for Elementary and Middle Schools

12/15/25

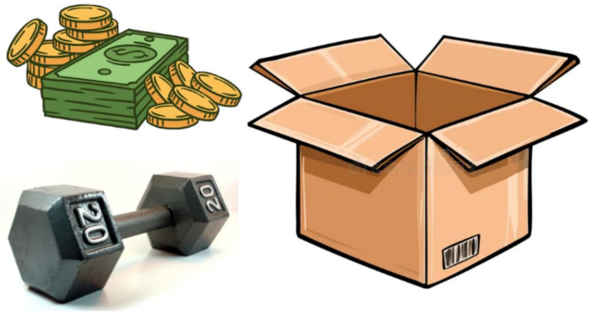


Strategy: Draw a diagram showing time.
Mark the opposite shores in minutes, 0 to 35. The slower boat touches alternating shores at 7, 14, 21, 28, and 35 minutes. The faster boat touches alternating shores at 5, 10, 15, 20, 25, 30, and 35 minutes. Where the paths cross, the boats are passing.

From the diagram, the faster boat passes the slower boat 7 times.

MOEMS®
Mathematical Olympiads for Elementary and Middle Schools

12/08/25

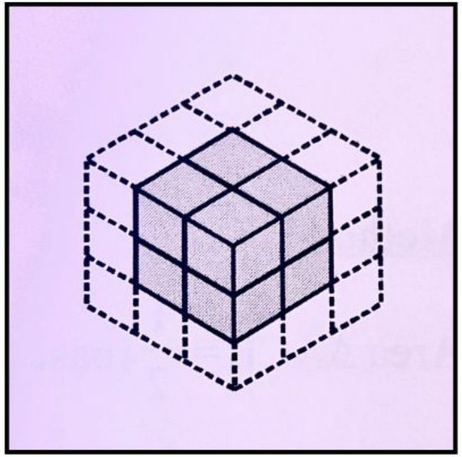


Strategy: Divide by the cost of the remaining pounds
Total Charge: $3.45
Charge for first five pounds: $1.65
Charge for remaining pounds: $1.80
Since the charge for each of the remaining pounds is 12¢ per pound, the number of remaining pounds is 1.80/12 = 15.

The total number of pounds in the package is 15 + 5 or 20.

MOEMS®
Mathematical Olympiads for Elementary and Middle Schools

12/01/25



Strategy: Use spatial reasoning and corner analysis
Only the 2 by 2 by 2 cubes at the eight vertices (corners) of the 3 by 3 by 3 cube have exactly three red faces.

The diagram below shows the location of one of the eight cubes.

MOEMS®
Mathematical Olympiads for Elementary and Middle Schools

11/24/25

Strategy: Use the work-rate method
In one minute, the water faucet fills 1/15 of the tub and the drain empties 1/20 of the tub. Together, the faucet and drain "fill" 1/15 - 1/20 of the tub each minute. This is equivalent to 4/60 - 3/60 or 1/60 of the tub.

Therefore, in 60 minutes the faucet and drain working together will fill 60/60 or the entire tub.

Volume 1, Olympiad 56, #4, 6 Minutes

MOEMS®
Mathematical Olympiads for
Elementary and Middle Schools

11/17/25

Strategy: Feed all the cats first.
One quart of milk feeds 6 cats so 15 cats require $2\frac{1}{2}$ quarts of milk. This leaves z quart of milk.

One quart of milk feeds 10 kittens, so 5 kittens can be fed with the leftover milk.

MOEMS®
Mathematical Olympiads for Elementary and Middle Schools

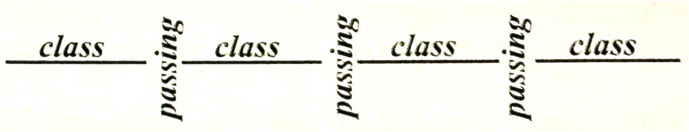
11/10/25

Strategy: Proceed one condition at a time.
The thousands digit can't be 2 or greater because then the ones digit would be more than 9. Thus, the thousands digit is 1. Also, the question reads, "Dr. Bolton WAS born..."
Condition:
The tens digit is twice the thousands digit ...... 1 _ 2 _
The ones digit is 3 times the tens digit ...... 1 _ 2 6
The hundreds digit is the sum of the other three digits 1 9 2 6

Dr. Bolton was born in 1926.

MOEMS®
Mathematical Olympiads for Elementary and Middle Schools

11/3/25



Strategy: Represent the problem visually.
The elapsed time from 8:26 to 11:26 is 3 hours, which is 180 minutes. There are 3 (not 4) passing periods between classes, for a total passing time of 12 minutes. This leaves a total of 168 minutes for the four classes.

One class period contains 42 minutes.

MOEMS®

Mathematical Olympiads for Elementary and Middle Schools

10/27/25

Strategy: Examine the differences between scores. The least sum is 3 x 2 = 6 and the greatest sum is 3 x 8 = 24. The 5-ring scores 3 points more than the 2-ring and the 8-circle scores 3 points more than the 5-ring. Thus "moving" a dart from the 2-ring increases the sum by a multiple of 3. Hence, all possible sums are multiples of 3 and are between 6 and 24, inclusive;

7 sums are possible.

MOEMS®
Mathematical Olympiads for Elementary and Middle Schools

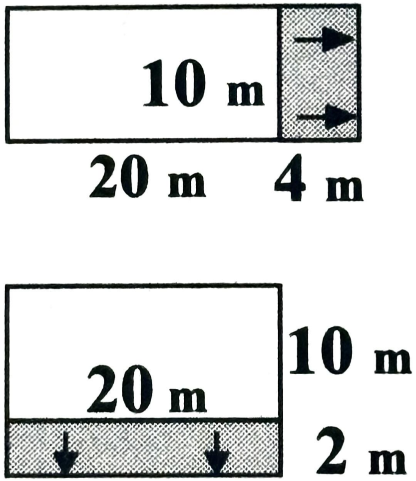
10/21/25

Strategy: Choose the gumballs so as to avoid three of a color. After Anna buys 8 gumballs, she might have 2 red, 2 green, 2 yellow, and 2 purple. The next gumball she buys must be one of those same four colors, guaranteeing her three of the same color. So, the minimum number of gumballs she should expect to buy is 9.

The minimum cost that guarantees three gumballs of the same color is $5 \times 9 = 45$¢.

MOEMS®
Mathematical Olympiads for Elementary and Middle Schools

10/13/25



Strategy: Draw a picture.
There are two possibilities:

1. In the first picture, the 10 m side has moved right. To enclose an extra 40 sq m, the two 20 m sides are each lengthened by 4 m.

This requires an extra 8m of fencing.

2. In the second picture, the 20 m side has moved down. To enclose an extra 40 sq m, the two 10m sides are each lengthened by 2 m. This requires an extra 4 m of fencing.

The least number of meters of additional fencing needed is 4 m.

MOEMS®
Mathematical Olympiads for Elementary and Middle Schools

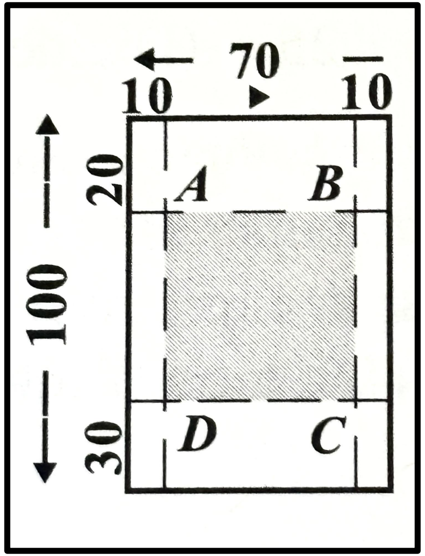
10/6/25

Strategy: Combine one stamp of each kind into a “super Stamp”.
Imagine one "super stamp" consisting of one of each of the stamps mentioned. This super stamp costs 50 + 20 + 10 + 5 = 85¢. There are 510 ÷ 85 = 6 of these super stamps.

Therefore Jessie has 6 fifty-cent stamps.
****This method illustrates the distributive property****

MOEMS®
Mathematical Olympiads for Elementary and Middle Schools

9/29/25



Strategy: Draw a diagram
As shown, the front of house ABCD would be 70 - 10 - 10 = 50 feet. The side of house ABCD would be 100 - 20 - 30 = 50 feet.

The largest area the house can have is 50 x 50 = 2500 sq. ft.

Volume 2, Set 5, Olympiad 5, 5D. 6 Minutes

MOEMS®

Mathematical Olympiads for Elementary and Middle Schools

9/22/25

Strategy: Find the total number of points earned.
In each race 5 + 3 + 1 = 9 points are earned. The three racers earn a total of 3 × 9 = 27 points. If each racer has the same score, each would have 9 points. Thus, Beth's total score is greater than
9. Each runner's total is the sum of three odd numbers and therefore is odd. The least odd number greater than 9 is 11, which is a possible winning score. For example, the scores could be 5, 5, 1; 3, 3, 3; 1, 1, 5.

Beth's least possible total score is 11.

Volume 2, Set 10, Olympiad 2, 2C. 6 Minutes